

Q1: 24 Feb (Shift 1) - Single Correct

If Y , K and η are the values of Young's modulus, bulk modulus and modulus of rigidity of any material respectively. Choose the correct relation for these parameters.

(1) $K = \frac{Y\eta}{9\eta - 3Y} \text{ N/m}^2$

(2) $\eta = \frac{3YK}{9K + Y} \text{ N/m}^2$

(3) $Y = \frac{9K\eta}{3K - \eta} \text{ N/m}^2$

(4) $Y = \frac{9K\eta}{2\eta + 3K} \text{ N/m}^2$

Q2: 24 Feb (Shift 2) - Numerical

A uniform metallic wire is elongated by 0.04 m when subjected to a linear force F . The elongation, if its length and diameter is doubled and subjected to the same force will be _____ cm.

Q3: 26 Feb (Shift 1) - Single Correct

The normal density of a material is ρ and its bulk modulus of elasticity is K . The magnitude of increase in density of material, when a pressure P is applied uniformly on all sides, will be :

(1) $\frac{\rho K}{P}$

(2) $\frac{K}{\rho P}$

(3) $\frac{PK}{\rho}$

(4) $\frac{\rho P}{K}$

Q4: 26 Feb (Shift 2) - Single Correct

The length of metallic wire is l_1 when tension in it is T_1 . It is l_2 when the tension is T_2 . The original length of the wire will be :

(1) $\frac{l_1 + l_2}{2}$

(2) $\frac{T_1 l_1 - T_2 l_2}{T_2 - T_1}$

(3) $\frac{T_2 l_1 + T_1 l_2}{T_1 + T_2}$

(4) $\frac{T_2 l_1 - T_1 l_2}{T_2 - T_1}$

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Answer Key

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Q1 (1)

Q2 (2)

Q3 (4)

Q4 (4)

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Q1 (1)

$$\Rightarrow y = 3k(1 - 2\sigma)$$

$$\Rightarrow \sigma = \frac{1}{2} \left(1 - \frac{y}{3k} \right) \dots (1)$$

$$\Rightarrow y = 2\eta(1 + \sigma)$$

$$\Rightarrow \sigma = \frac{y}{2\eta} - 1 \dots (2)$$

by comparing equation (1) and (2), we get

$$\Rightarrow \frac{y}{2\eta} - 1 = \frac{1}{2} \left(1 - \frac{y}{3k} \right)$$

$$\Rightarrow \frac{y}{\eta} - 2 = 1 - \frac{y}{3k}$$

$$\Rightarrow \frac{y}{\eta} = 1 + 2 - \frac{y}{3k} \Rightarrow \frac{y}{\eta} = 3 - \frac{y}{3k}$$

$$\Rightarrow \frac{y}{3k} = 3 - \frac{y}{\eta} \Rightarrow \frac{y}{3k} = \frac{3\eta - y}{\eta}$$

$$\Rightarrow k = \frac{\eta y}{9\eta - 3y}$$

Q2 (2)



$$y = \frac{F/A}{\Delta\ell/\ell}$$

$$\Rightarrow \frac{F}{A} = y \frac{\Delta\ell}{\ell}$$

$$\Rightarrow \frac{F}{A} = y \times \frac{0.04}{\ell} \dots (1)$$

When length & diameter is doubled.

$$\Rightarrow \frac{F}{4A} = y \times \frac{\Delta\ell}{2\ell} \dots (2)$$

$$(1) \div (2)$$

$$\frac{F/A}{F/4A} = \frac{y \times \frac{0.04}{\ell}}{y \times \frac{\Delta\ell}{2\ell}}$$

$$4 = \frac{0.04 \times 2}{\Delta\ell}$$

$$\Delta\ell = 0.02$$

$$\Delta\ell = 2 \times 10^{-2}$$

$$\therefore x = 2$$

Q3 (4)

$$\text{Bulk modulus } K = \frac{-\Delta P}{\frac{\Delta V}{V}} = \frac{-\Delta P V}{\Delta V}$$

$$\text{We know, } \rho = \frac{M}{V}$$

$$\text{So, } \frac{-\Delta \rho}{\rho} = \frac{\Delta V}{V}$$

$$K = \frac{-\Delta P}{\left(\frac{-\Delta \rho}{\rho}\right)} = \frac{\rho \Delta P}{\Delta \rho}$$

$$\Delta \rho = \frac{\rho \Delta P}{K}$$

$$\Delta \rho = \frac{\rho P}{K}$$

Q4 (4)

$$\text{From young's modulus relation } \left(y = \frac{\frac{F}{A}}{\left(\frac{\Delta l}{l}\right)} \right)$$

we can write for 1st case

$$\frac{T_1}{A} = \frac{y(\ell_1 - \ell)}{\ell}$$

we can write for 2nd case

$$\frac{T_2}{A} = \frac{y(\ell_2 - \ell)}{\ell}$$

$$\frac{T_1}{T_2} = \frac{\ell_1 - \ell}{\ell_2 - \ell}$$

$$T_1 \ell_2 - T_1 \ell = T_2 \ell_1 - T_2 \ell$$

$$\frac{T_2 \ell_1 - T_1 \ell_2}{T_2 - T_1} = \ell$$